The isotropic Schwarzschild relation ψ + Nψ = 2: a counting reading with one transfer postulate, and a closure inside matter

Ugo Lanciano
Independent researcher, Locri, Italy · ORCID 0009-0008-7618-9635

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Abstract

In isotropic coordinates the Schwarzschild metric satisfies ψ + Nψ = 2 for every mass, where N is the lapse and ψ the conformal factor of space. This record reads the two numbers of that relation from count vectors ("registers") with Dirichlet posteriors, and keeps apart what is computed, what is a premise of the author's framework, what is a choice, and one postulate. With five stated premises, and in the limit of many readings (corrections of order 1/T, measured): the lapse is the ratio of the Fisher steps per reading of two places, $N = \tau_0/\tau_x$; the depth is the normalised overlap of the two descriptions projected on the plane perpendicular to the line of sight, $\psi = 2\tau_x/(\tau_x + \tau_0)$; ψ/2 and Nψ/2 are then the shares of the place and of the distant observer in their joint count, and ψ + Nψ = 2 says that the two shares sum to one. The fragmentation of a register behaves as an isotropic Brownian motion in the three-dimensional Fisher space and deposits a measure 1/(2πr) − B with no free parameter. The conversion from that deposit to ψ is stated as a postulate (transfer of share, $\psi - 1 = \Sigma\, Gm/(2rc^2)$), chosen among four normalisations because it reproduces general relativity exactly: two alternatives are excluded by planetary ephemerides (β = 3/2 and 1/2), one differs from Schwarzschild only at third order. With it, one body gives the isotropic Schwarzschild metric (γ = β = 1) and two bodies …

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Version 2.36.2 adds section 11: the relation ψ + Nψ = 2 read as an exchange that keeps the distant observer's unit fixed, the second order it fixes, and the deposit of 3.6 read as a probability of arrival, with the registered tests of a fragmentation bench (`documenti/ARRIVAL_TESTS.md`), nine programs and their outputs; `codice/run_all.py` runs them too. It also gives the precise reference for the normalisation −Φ/2c² (sections 3.7 and 9): it is that of the preprint of Verlinde's paper, while the published version uses −2Φ/c². Version 2.36.1 added sections 9 and 10. Every other program, result and log is unchanged.

MM is the author's framework in which physical quantities are read off from count vectors ("registers") compared through Bayesian overlaps; this deposit uses only the small part of it defined in "The model in five lines" below.

1. The identity (outside matter)

Write a static, spherically symmetric metric in isotropic coordinates, with spatial part $\psi^4(dr^2 + r^2 d\Omega^2)$ and lapse N (G = c = 1). In vacuum general relativity ψ = 1 + M/2r and Nψ = 1 − M/2r are both harmonic, so

ψ + Nψ = 2

holds exactly for the Schwarzschild metric, with equal monopoles, M_ADM = M_Komar = M.

The converse is checked for **one centre** in closed form (`codice/einstein_vuoto_esatto.py`): ψ = 1 + a/r with N = 2/ψ − 1 gives the isotropic Schwarzschild metric with M = 2a, so the exterior has γ = β = 1. It does not extend to non-spherical harmonic ψ: for $\psi = 1 + a_1/|x - x_1| + a_2/|x - x_2|$ the Ricci tensor of the static metric is non-zero at order $a_1 a_2$ in an open region, and vanishes when either mass is removed (`codice/domini_e_masse.py`). (This ψ is the Brill–Lindquist conformal factor of time-symmetric vacuum data; the failure here concerns the static lapse N = 2/ψ − 1.) The relation therefore holds only in exact spherical symmetry. **Domain of this deposit: static, spherically symmetric, one body.**

2. The model in five lines

1. A **register** is a count vector $c \in \mathbb{N}^K$ with K = 4 and $c_k \geq 1$, total n; its state is the Dirichlet posterior with evidence $E(c) = \log[(K-1)!\, \prod_k c_k! / (\Sigma_k c_k + K - 1)!]$.

2. Registers are sampled **uniformly in the Fisher measure** on the simplex (Amari and Nagaoka 2000) ($p = g^2/|g|^2$ with g Gaussian), then c = 1 + Multinomial(n − K, p). Distances are Fisher distances, d(p, p′) = 2 arccos Σ_k √(p_k p′_k). Taking statistical distinguishability as a distance goes back to Wootters (1981); information geometry as the geometry of space is Caticha's (2005, 2016, 2019; see section 9).
3. Two registers are linked with the **normalised Bayesian overlap** w_ij = exp[E(c_i + c_j) − ½E(2c_i) − ½E(2c_j)], self-weight 1, keeping links within 3.5 widths (width $2/\sqrt{n}$ in the coordinates $2\sqrt{(c/n)}$).
4. The operator is the **normalised graph Laplacian** L = I − D^(−1/2) a D^(−1/2), with $a = d^{-1}Wd^{-1}$, d the row sums of W and D the row sums of a.
5. The **static response** to a point source placed at the most uniform register solves L G = δ inside a Fisher ball of radius 0.7 around it (G = 0 outside; the radius is a numerical choice); the potential is φ = D^(−1/2) G.

The graph lives in the three-dimensional space of register states. Identifying distances in that space with physical distances is a premise of the framework (P3 in section 3), not a derived result.

Section 3 adds three objects of the same framework: the effective density of a register, the image of a register seen along a line, and the fragmentation of a register. Section 7 uses a separate bench, stated there. The code of the graph is `codice/grafo_uno_su_r.py`.

3. From counting premises to ψ and N

This section reads the two numbers of the relation from registers. It keeps apart what is computed [C], what is a premise of the framework [P], what is a choice among alternatives [S], and one postulate [T]. The chain is static; its laws hold in the limit of many readings, with corrections of order 1/T.

3.1 The elementary identity. [C] For two isotropic Gaussians p_0, p_x in D dimensions with a common centre and variances s_0^2 (the distant observer's resolution) and s_x^2 (the resolution at the place), let Λ_ab = ∫ p_a p_b, ℓ(a → b) = Λ_ab/Λ_bb and $q = s_0^2/s_x^2$. Then ℓ(0 → x) = (2/(1+q))^(D/2) and ℓ(x → 0) = (2q/(1+q))^(D/2), and in D = 2 the two overlaps sum to 2 for every q. With the relative entropy the two directions do not add up to a constant: KL(0‖x) + KL(x‖0) = (D/2)(q − 1)²/q (`codice/reciprocita_nello_specchio.py`). The rest of the section says why the two overlaps are the depth ψ and the product Nψ.

3.2 Premises. [P] They are the axioms of the framework, stated here in technical form.

- **P1 (clock).** The proper time of a place advances, per reading, by the Fisher length of the update of its register; readings arrive at the same rate everywhere. (Time measured by the size of fluctuations is the entropic time of Caticha (2011), made local by Ipek, Abedi and Caticha (2018) and Ipek and Caticha (2020). There the lapse is given and the fluctuations are set by it; here the direction is reversed: see section 9.)
- **P2 (image).** Two points of view describe the same thing (same composition) with different numbers of readings, and compare their descriptions through the image perpendicular to the line joining them: a two-dimensional comparison.
- **P3 (distance).** Distance is the measure of the fragmentation of a register: the displacement of its copies in the space of register states.
- **P4 (mass).** The number of copies of a register (the multiple) is mass; its effect adds.
- **P5 (absorption).** A copy that reaches a place is absorbed into the joint state of the two points of view, which stays complete.

3.3 The clock: $N = \tau_0/\tau_x$. [C] With the effective density τ = T + 1 (T readings, prior pseudo-counts included), the root-mean-square Fisher step of one new reading, in the local Fisher metric, is $\sqrt{(K-1)/\tau}$ for every composition of the register (symbolic for K = 4; numerically within 0.23 %, equal across compositions within 0.35 %; the finite Fisher–Rao distance agrees when every category holds many readings). The ratio of the steps of two places is τ_0/τ_x (within 0.3 %), and for equal compositions the Dirichlet posterior variances give the same ratio exactly (`codice/ritmo_per_lettura.py`, `codice/t1_densita.py`, `codice/t5_coerenza.py`). With P1 the lapse, the rate of the place's clock against the distant one, is this ratio: $\mathbf{N = \tau_0/\tau_x}$. The form of this statement, the lapse as the ratio of an intensive quantity far away to the same quantity at the place, is that of the Tolman–Ehrenfest relation, T_loc N = T_∞ (Tolman and Ehrenfest 1930), with the density of readings in place of the temperature.

3.4 The depth: $\psi = 2\tau_x/(\tau_x + \tau_0)$. [C, many-readings limit] Project the posteriors Dir(τp), in Fisher coordinates y = 2√p, onto the plane perpendicular to the line of sight (P2), and take the normalised overlap seen from the place, L⊥(x → 0) = ∫f_x f_0 / ∫f_0^2. Computed on the true projected images (sampled, with no Gaussian formula), it equals $2\tau_x/(\tau_x + \tau_0)$ within 0.44 % at T_0 = 100, 0.10 % at 300 and 0.03-0.04 % at 1000-3000, a correction of order 1/T, for any direction of the line within 0.1 % (`codice/t9_immagine_esatta.py`; control of the estimator on discs and Gaussians in `codice/t9c_controllo.py`). The ratio of the two directions, $\int f_0^2/\int f_x^2$, tends to τ_0/τ_x: the same N as in 3.3, by a

second route. In the full three-dimensional state space the overlap goes instead as the 3/2 power (`codice/t6_specchio.py`). [S] The overlap seen from the place is identified with ψ. Any comparison symmetric in the two points of view is even in $z = (\tau_x - \tau_0)/(\tau_x + \tau_0)$ and, identified with ψ, gives γ = 0; the directional one is chosen, which responds linearly to the signed excess of readings.

3.5 Two shares of one joint count. [C] In the limit of 3.4, $\psi/2 = \tau_x/(\tau_x + \tau_0)$ and $N\psi/2 = \tau_0/(\tau_x + \tau_0)$: the shares of the place and of the distant observer in the joint count of the two points of view. **$\psi + N\psi = 2$ says that the two shares sum to one.** This is the reading proposed for the relation, outside matter and inside (section 4); section 11 reads it as the conservation law of an exchange. In isotropic coordinates the conformal geometry of space is flat and all the content of the spatial metric is in ψ: the local scale that information geometry leaves free (Caticha 2016; section 9).

3.6 The source: fragmentation gives 1/r. [C] The fragmentation of a register with K = 4, its copies losing precision reading by reading, behaves as an isotropic Brownian motion in the three-dimensional Fisher space, with clock s equal to the sum of the squared Fisher steps (60 000 chains: isotropic within 1.8-5 %, variance per direction equal to s within 1.4 %). Weighting the deposit by the precision lost, Δs, the measure deposited at distance r is $1/(2\pi r) - B$ with no free parameter within its conventions (Fisher coordinates, clock s, tangent Euclidean volume, cut-off n_min; it does not give G in physical units): mean ratio 0.998, no class beyond 3 %, for r between 0.005 and 0.04; B comes from the end of the fragmentation, and the whole profile is $[1/(2\pi r)]\ \mathrm{erfc}(r/\sqrt{(2s_{max})})$ (`codice/t8b_frazionamento.py`). This is the Green function of Brownian motion in three dimensions; what the framework adds is that the fragmentation of a K = 4 register behaves as such a motion, with that step. [S] The loss mode and the weight Δs are choices (with a weight per reading the registered slope was missed). With P3, r is the physical distance; with P4, n copies deposit n times.

3.7 The junction: one postulate. Nothing in 3.3-3.6 converts the deposited measure into ψ. The conversion is stated as a postulate.

> **Postulate T (transfer).** Each deposit moves share from the distant observer to the place: $\psi - 1 = \Sigma_i w_i$, with $w_i = Gm_i/(2r_ic^2)$. The coefficient is calibrated (G is measured), and the shares stay positive ($\Sigma_i w_i < 1$; for one body, $r > GM/(2c^2)$).

T is motivated by P5, since an absorbed copy re-divides a complete joint state, but it is not implied by it: adding counts and renormalising also gives a complete state. The quantity that T moves, $w = GM/(2rc^2)$ in the isotropic radius, coincides with $-\Phi/2c^2$ at first order in the weak field. In the preprint of Verlinde's paper (arXiv:1001.0785v1, 2010, eq. 3.16 and section 4) this quantity is the amount of coarse graining per bit, between 0 and 1 and equal to 1 only on a black-hole horizon; the published version (JHEP 04 (2011) 029, eq. 3.12 and section 4) writes the same statement as $-2\Phi/c^2$, which equals 1 on the horizon in areal coordinates, $r = 2GM/c^2$. In isotropic coordinates the horizon lies at $r = GM/(2c^2)$, where w reaches 1 ($\psi = 2$, $N = 0$); this value comes from writing T in the isotropic radius, not from an identity with Verlinde's quantity beyond first order. It is chosen among four normalisations of the excess of readings, each set proportional to $u = GM/(rc^2)$ with the Newtonian limit fixed (`codice/normalizzazioni.py`):

excess normalised to	β	γ	third order of N	against data
S1 the distant count, $(\tau_x - \tau_0)/\tau_0$	3/2	1	$-u^3$	excluded: $\beta - 1$ is below about 10^{-4} (Will 2014)
S2 the place, $(\tau_x - \tau_0)/\tau_x$	1/2	1	0	excluded
S3 the geometric mean, $(\tau_x - \tau_0)/\sqrt{(\tau_x\tau_0)}$	1	1	$-u^3/8$	passes; differs from Schwarzschild at third order
S4 the joint count, $(\tau_x - \tau_0)/(\tau_x + \tau_0) = \psi - 1$	1	1	$-u^3/4$	isotropic Schwarzschild at every order

S4 is chosen because it reproduces general relativity exactly: an external criterion of correspondence, not a selection by the premises. S3 would differ only in strong fields.

3.8 What follows with T. For one body $\psi = 1 + GM/(2rc^2)$ and, by 3.5, $N = 2/\psi - 1$: the isotropic Schwarzschild metric exactly, $\gamma = \beta = 1$ (`codice/einstein_vuoto_esatto.py`, `codice/perielio.py`). For two bodies $\psi = 1 + \Sigma_i Gm_i/(2r_ic^2)$ satisfies the vacuum Hamiltonian constraint: the time-symmetric initial data of Brill and Lindquist (1963); with counts added (S1) the constraint fails (`codice/t11_due_masse.py`). The lapse $2/\psi - 1$ extended to two poles does not satisfy the static equations, $E_{ij} = R_{ij} - D_iD_jN/N \neq 0$ (`codice/t11b_tensore.py`, `codice/domini_e_masse.py`): with this lapse, two bodies at rest are initial data that evolve, as in general relativity; E_{ij} is not a measured force, and the chain contains no dynamics.

3.9 Status. Schwarzschild is not derived here: it enters through Postulate T, with the 1/r of 3.6, and the choice S4. What the counting chain supplies is the form of the two quantities (a ratio of densities and a share), the relation between them as the completeness of two shares, and the 1/r of the deposit. The registered tests, including the predictions that failed, are listed in `documenti/TUNNEL_TESTS.md`.

4. Inside matter: a phenomenological closure, not yet a theory

In the reading of section 3 the relation says that the two shares of the joint count sum to one (3.5). Keeping it inside matter, as below, means keeping this completeness point by point.

In general relativity, inside a static star in isotropic coordinates, the Hamiltonian constraint is $\nabla^2\psi = -2\pi\rho\psi^5$ and the lapse equation is $\nabla^2(N\psi) = 2\pi N\psi^5(\rho + 6p)$, so

$$\nabla^2(\psi + N\psi) = 2\pi\psi^5[N(\rho + 6p) - \rho].$$

The right-hand side is of first post-Newtonian order. Its volume integral is a surface flux: with $\psi \to 1 + M_{ADM}/2r$ and $N \to 1 - M_{Komar}/r$,

$$\int \nabla^2(\psi + N\psi)\, d^3x = 4\pi(M_{Komar} - M_{ADM}),$$

which vanishes for a static star in general relativity, where Komar and ADM masses coincide for stationary asymptotically flat spacetimes (Beig 1978; see also Tolman 1930). The relation, kept inside matter, imposes point by point what general relativity guarantees only globally (`codice/domini_e_masse.py`).

The choice of variable. The relation is stated for the isotropic conformal factor ψ, i.e. with the spatial metric written as $\psi^4\delta_{ij}$ (conformally flat). This choice is an assumption. In areal coordinates the analogous vacuum statement is $g_{tt}\, g_{rr} = -1$ (Jacobson 2007), and imposing that inside matter would give a different closure.

What is replaced, and what it costs. The model keeps the Hamiltonian constraint, replaces the lapse equation by $N = 2/\psi - 1$, and takes the pressure from energy-momentum conservation, $dp/dr = -(\rho + p)\, d\ln N/dr$. The price is that the fluid's pressure does not enter the field equations: inside matter the metric is no longer generated by the fluid through Einstein's equations. This is a phenomenological closure of the static equations, not yet a theory. For a uniform-density star (`codice/interno_uniforme.py`), $-G^t_t/8\pi = \rho$ exactly, but the spatial components differ from the fluid pressure: at the centre the geometry sees about half of it ($G^r_r/8\pi = G^\theta_\theta/8\pi \approx 0.50\, p_c$ in the weak field), and towards the surface the difference becomes anisotropic, with a positive tangential part. Read in general relativity, the star carries an extra, conserved, zero-energy anisotropic stress. Buchdahl's 4/9 assumes an isotropic perfect fluid with density non-increasing outwards; Andréasson (2008) proves $2m/r \le 8/9$ for any static spherical matter with $p_r \ge 0$ and $p_r + 2p_t \le \rho$, anisotropic or not (anisotropic spheres: Bowers and Liang 1974). A star above 4/9, read in general relativity, must therefore violate one of those two conditions through the extra stress. Inside matter this is not general relativity plus an assumption.

Closed form of the bound. For uniform proper density the Hamiltonian constraint is the Lane–Emden equation with $n = 5$, solved by $\psi = \psi_c(1 + br^2)^{(-1/2)}$. Matching to the exterior gives $\psi_R = \psi_c^{(2/3)}$. The sequence ends when the centre saturates, $\psi_c = 2$ ($N_c = 0$), at

$$C_{max} = GM/(Rc^2) = 2^{(1/3)} - 2^{(-1/3)} \approx 0.4662,$$ against Buchdahl's $4/9 \approx 0.4444$.

At observed compactness. For uniform density, at $C = 0.16$, 0.19 and 0.22 the central pressure needed by the model is lower than in general relativity by 7.4 %, 9.3 % and 11.4 %. Uniform density is not realistic; with a realistic equation of state the mass–radius curve would shift, and computing that shift is the next step, not done here. Several density profiles and the energy conditions are treated in the deposit MM 2.34 (same author, Zenodo).

5. Assumptions

#	assumption	status	what would refute it
1	Source: $\psi - 1 = \Sigma_i w_i$, $w_i = Gm_i/(2r_ic^2)$ (Postulate T, section 3.7)	postulate, chosen among four normalisations for exact agreement with general relativity	a strong-field measurement favouring S3, which changes the metric at third order
2	Harmonic propagation: the deposit falls as	the fragmentation (3.6) and the graph (6) give 1/r in the space of register states; its	the free exponent stopping below 1 at finer

	1/r	identification with physical space is premise P3	resolution and with more seeds
3	The relation $\psi + N\psi = 2$ everywhere, including inside matter, where it replaces the lapse equation of general relativity	assumed (in the reading of 3.5: the completeness of the two shares, kept point by point; in the reading of 11.2: the conservation law of an exchange); outside matter it is a property of general relativity	a mass–radius measurement favouring general relativity once a realistic equation of state is used (0.466 is the maximum for uniform density only; for densities decreasing outwards the maximum is lower, MM 2.34)
4	Boundary condition $\psi \to 1$ at infinity (the distant observer's unit)	assumed	— (with $\psi = 3/2 + a/r$ the Einstein tensor is not zero)
4′	The variable is the isotropic conformal factor: spatial metric $\psi^4\delta_{ij}$	assumed (in areal coordinates the analogue is $g_{tt}\, g_{rr} = -1$, Jacobson 2007)	—
5	Metric form $g_{tt} = -N^2$, $g_{ij} = \psi^4\delta_{ij}$; test particles follow geodesics, $H = N\sqrt{(m^2 + p^2/\psi^4)}$	assumed	an observable of time or distance that does not follow this form
6	Universality of the lapse (equivalence principle)	assumed	a free fall that depends on the material (today universal to 10^{-15})
7	G is calibrated by a measurement; that a is proportional to the mass is assumed	assumed	a G that differs from body to body
P1-P5	Premises of the framework (section 3.2)	axioms	—
P6-P7	Premises of the moving clock (section 10): second link, no pre-assigned places	axioms	—
P8	Premise of the exchange (section 11.2)	axiom	—
—	Section 11.2: the two shares as the exchanging parts, their sum as the conserved quantity, light reading the product of the local units of time and length	declared formalisation of P8	—
—	Moving clock: the clock reads the number of copies and combines opposite light directions by product (section 10)	declared choice	—
—	Choices: effective density $\tau = T + 1$; the directional overlap identified with ψ; loss mode and weight Δs of the fragmentation; same composition of the two points of view	declared choices (3.3-3.6)	—
—	Many-readings limit: the laws of 3.3-3.5 hold with corrections of order 1/T	measured: 0.44 % at T = 100, 0.03 % at 3000	—

6. The 1/r on the model's graphs

The free exponent p of A r^(−p) + B, fitted to shell averages of φ, is 0.937, 0.956 and 0.980 for n = 512, 1024 and 2048 counts per register (64 000, 181 019 and 512 000 registers; three points, one seed each, no error bars). This is expected: graph Laplacians built with Gaussian-like kernels converge to the Laplace–Beltrami operator of the underlying space (Belkin and Niyogi 2008; Coifman and Lafon 2006), and the space of register states for K = 4 is three-dimensional. **The 1/r reflects K = 4; it is not a dynamical result.** Section 3.6 gives a second route to 1/r, from the fragmentation of one register; both rest on the three dimensions of the K = 4 state space and hold in that space, which premise P3 identifies with physical space. The code is included (`codice/grafo_uno_su_r.py`); n = 512 runs on a laptop in about a minute, while n = 1024 and 2048 need more memory than 8 GB (n = 2048 took 135 s on a 128 GB machine).

7. The parameter γ

The two statements below concern different objects. For the continuum exterior metric γ = 1 exactly (section 1). On a

separate finite-count bench (a ring of registers with squared overlaps), and granting the link $\gamma = 2p - 1$ from the author's working files, which is not part of this deposit and cannot be checked here, one finds $\gamma - 1 \approx 7.9/n$, where p is the exponent with which the lowest non-zero mode of the graph scales with the resolution. Keeping $\gamma - 1$ below the Cassini uncertainty, 2.3×10^{-5} ($\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$, Bertotti, Iess and Tortora 2003), then requires $n > 3.4 \times 10^5$ counts per register. This was found after the registered test (`documenti/NOTE_GAMMA.md`, `codice/gamma_anello.py`).

8. What is not claimed

- That Schwarzschild is derived: outside matter the relation is a property of general relativity; in the chain of section 3 it enters through Postulate T and the choice S4.
- That the premises P1-P5 are results: they are the axioms of the framework, and the choices of section 3 are declared.
- That the laws of 3.3-3.5 hold exactly at a finite number of readings: they hold in the limit, with corrections of order 1/T.
- That the 1/r follows from a dynamics: it is the Green function of a Brownian motion (3.6) and the expected behaviour of the graph Laplacian in three dimensions (6).
- That G is predicted: it is measured.
- That 0.466 bounds every density profile: it is the maximum for uniform density.
- That the model reproduces general relativity inside matter: it does not, by construction; inside matter it is a phenomenological closure, not yet a theory.
- A dynamics or a covariant form: none is proposed; two bodies at rest are initial data, not a static solution.
- That the ideas listed in section 9 as prior work are new: they are not; what this deposit adds is stated there.
- That the moving clock of section 10 is new: as mathematics it is causal-set theory and Bondi's k-calculus; nor that it is joined to the lapse of section 3.
- That section 11 derives $\psi + N\psi = 2$, Postulate T, G or the radius $GM/(2c^2)$: it gives the relation a reading and the deposit a probabilistic meaning. That the fragmentation bench proves the Brownian limit of the first arrival: its results are compatible with it.

Other lines of work are not part of this deposit.

9. Relation to prior work

The chain of section 3 uses ideas that are not new. Each is listed with what the source does and with what differs here.

element of this deposit	prior work	what differs here
distance as Fisher distinguishability (section 2, P3)	Wootters (1981): statistical distance. Caticha (2005, 2016, 2019): the metric of space is the information metric of a blurred point; it fixes the conformal geometry of space but not its local scale (2016); with the blur as unit and the embeddability conditions of Hojman, Kuchař and Teitelboim, the vacuum Einstein equations follow (2019)	the statistical object is a count register with its Dirichlet posterior, not the blur of a position; the three dimensions come from $K = 4$
P1: proper time advances with the size of a fluctuation	entropic time, Caticha (2011); local entropic time, Ipek, Abedi and Caticha (2018), Ipek and Caticha (2020): at each point the variance of the fluctuations grows by equal amounts in equal proper times $\delta\xi_\perp = N\,dt$	there the lapse is given by the geometry and the fluctuations are set by it; here the step per reading depends on the density of readings, $\sqrt{(K-1)}/\tau$, and the ratio of two steps gives the lapse
$N = \tau_0/\tau_x$, a ratio of the far to the near	Tolman and Ehrenfest (1930): $T_{loc}\,N = T_\infty$	the intensive quantity is the density of readings
ψ, the conformal factor	Caticha (2016): the local scale left free by information geometry; in general relativity it is fixed by the Lichnerowicz–York equation, which for conformally flat, time-symmetric vacuum data reduces to $\nabla^2\psi = 0$	ψ is read as a share of a joint count (3.4-3.5); with Postulate T, $\psi = 1 + \Sigma_i\, Gm_i/(2r_ic^2)$, the Brill–Lindquist solution of that equation
Postulate T: w, equal to $-\Phi/2c^2$ at first order, as a share between 0 and 1	Verlinde: $-\Phi/2c^2$ as the amount of coarse graining per bit, between 0 and 1, equal to 1 only on a horizon (preprint, 2010); $-2\Phi/c^2$, equal to 1 on the horizon in areal coordinates (published version, 2011)	the completeness of two shares, $\psi + N\psi = 2$, which ties the lapse to the conformal factor; the exact isotropic Schwarzschild metric
counting degrees of freedom; gravity as thermodynamics	Padmanabhan (2010): $E = \tfrac{1}{2}\,k_B \int dn\, T_{loc}$, with $dn = \sqrt{\sigma}\,d^2x/L_P^2$ and $T_{loc} = N\,a\cdot n/2\pi$. Jacobson (1995, 2016): the Einstein equation as an equation of state, and from entanglement equilibrium	no reading of N or ψ as counts

In the works listed above we did not find: the lapse read as a ratio of densities of readings, with the Fisher step $\sqrt{(K -$

1)/τ; ψ read as the directional overlap of projected images, $2\tau_x/(\tau_x + \tau_0)$, with $\psi + N\psi = 2$ as the completeness of two shares; the fragmentation of a register, which behaves as a Brownian motion and deposits 1/r; the closure inside matter (section 4). These are what this deposit proposes, within the limits stated in sections 3.9, 5 and 8. The check covered the works listed here, not the whole literature.

10. A moving clock (outlook; added in version 2.36.1)

Section 3 concerns clocks at rest in a static field. For a clock in uniform motion in flat spacetime, the counting reading gives special-relativistic time dilation. Between the start and end events of the clock, light rays bound a diamond with sides $a = c\Delta t - \Delta x$ and $b = c\Delta t + \Delta x$. Two further premises of the framework enter, in technical form:

- **P6 (second link).** In an elementary step the two light-side counts are coprime. Copies are generated at their common multiples, so a diamond of sides a and b holds $ab = \gcd(a, b)\cdot\mathrm{lcm}(a, b)$ cells.
- **P7 (no pre-assigned places).** Places do not exist before the copies. A set of places fixed in advance would define a preferred frame. In the tests the copies are taken as independent and uniformly random in spacetime volume: an assumption that P7 motivates but does not imply.

[S] The clock reads the **number** of copies and combines the two opposite light directions by product. Its time is $\sqrt{(ab)}$. The laboratory time is $(a + b)/2$.

[C] Results (`documenti/MOVING_CLOCK_TESTS.md`; predictions hashed before computing):

arrangement of copies	what the clock reads	result
uniform at random, 1+1, 200 000 copies	longest chain	$\sqrt{(1 - v^2/c^2)}$ within +0.2 to −0.9 % for v = 0.3-0.9, and within 0.3 % after a finite-size correction
uniform at random, 3+1, 6000 copies	longest chain	a moving clock and a clock at rest with the same proper time agree within 1.3 σ; the fixed 6 % tolerance failed at v = 0.8 (about 780 copies)
synchronous grid	longest chain	$(a + b)/2$ for every v: Galilean time
fixed places, random times	longest chain	dilation 7-29 % stronger than Lorentz: the places define a frame
synchronous grid	number of copies, product rule	$\sqrt{(1 - v^2/c^2)}$ within 0.43 % (T = 1000)
light lattice	$\sqrt{(ab)}$ against $(a + b)/2$	$\sqrt{(1 - v^2/c^2)}$ to 3×10^{-16}

The reason fits in one line. The number of copies in the diamond is proportional to ab, so a clock that counts them, or the longest chain, which grows as the square root of the count, gives the geometric mean of the two sides. A clock that adds the sides gives the arithmetic mean. **The arithmetic mean exceeds the geometric mean except at rest (a = b), and the gap is time dilation:** $\sqrt{(ab)}/((a + b)/2) = \sqrt{(1 - v^2/c^2)}$.

A clock at rest steps by (1, 1). A step with coprime sides is indivisible: the segment from (0, 0) to (a, b) has $\gcd(a, b) - 1$ interior lattice points (no exception in 3600 pairs). A moving clock moves $(b - a)/2$ per step.

The success criterion registered for M1-M5 as a whole (if M1 or M5 failed, the dilation would not follow from the count) was not met, because M5 failed its 6 % tolerance at v = 0.8; the equal-proper-time comparison passed, and the finite-size reading was made afterwards (`documenti/MOVING_CLOCK_TESTS.md`).

Prior work. None of this is new as mathematics:

- Proper time as the longest chain of a Poisson sprinkling, and its Lorentz invariance: Brightwell and Gregory (1991); Bombelli, Henson and Sorkin (2009).
- In 1+1 the longest chain is Ulam's problem (Hammersley 1972; Vershik and Kerov 1977; Logan and Shepp 1977).
- Proper time as the geometric mean of the two light times: Bondi's k-calculus (Bondi 1964).

What the framework supplies is the reason for the arrangement. With P7 a lattice of places, which gives Galilean time, is excluded. With P6 the count is a product.

Not done here:

- Joining this count with the lapse N of section 3 in one chain. Gravitational and kinematic dilation are computed here by different counts.
- Curved spacetime.
- Correlated placements.

The continuation is the subject of a separate deposit in preparation: weights on the turns, which give the Feynman checkerboard and the Dirac equation, mass as the turning angle, and the field of MM 2.27 on the same lattice.

11. The relation as an exchange, and the deposit as a probability of arrival (added in version 2.36.2)

Section 3 reads $\psi + N\psi = 2$ as the completeness of two shares and uses it, with Postulate T, to fix the lapse. This section shows what the relation fixes at second order, reads it as an exchange that keeps the distant observer's unit fixed, and gives the deposit of 3.6 a second meaning: in the ideal Brownian model, with the target fixed by Postulate T, a probability of arrival.

11.1 What the relation fixes. [C] With $\psi = 1 + w$ and $w = GM/(2rc^2)$ from Postulate T, write the lapse as $N\psi^2 = F(w)$, the product of the local unit of time N and the local unit of length ψ^2, both relative to the distant observer. Reading the mass from g_tt at first order (`codice/ppn_quadrato.py`):

F(w)	$-g_{tt} = N^2$	γ	β	
1	$1 - 2U + 5U^2/2$	1	5/4	excluded by perihelia
$1 - w$	$1 - 3U + 19U^2/4$, mass rescaled	2/3	19/18	excluded by light deflection and Shapiro delay
$1 - w^2$	$1 - 2U + 2U^2$	1	1	isotropic Schwarzschild, $N = (1 - w)/(1 + w)$

$U = GM/(rc^2)$. In general, for $F = 1 + f_1 w + f_2 w^2 + \ldots$, $\gamma = 1/(1 - f_1/2)$ and, with $f_1 = 0$, $\beta = (5 + f_2)/4$: exact agreement with the Schwarzschild values $\gamma = \beta = 1$ requires $f_1 = 0$ and $f_2 = -1$, and the measurements bound the deviations (Will 2014; Bertotti, Iess and Tortora 2003 for γ). $\psi + N\psi = 2$ is exactly $F = 1 - w^2$, since $N\psi = 2 - \psi = 1 - w$; that it holds at every order is the content of S4 (3.7). The product also has a geometric meaning: with the areal radius $R = r\psi^2$, $dR/dr = 1 - w^2 = N\psi^2$ (`codice/quota_e_arrivo.py`).

11.2 The exchange. [P] **P8 (exchange).** Light is a change in the configuration of the description. The reference observer's unit of reading is stable; to keep it so, fractional parts of the description continuously exchange quanta, which are fractions of that unit.

[S] Formalised here by three identifications: the two exchanging parts are the shares of 3.5, $\psi/2 = (1 + w)/2$ and $N\psi/2 = (1 - w)/2$, which move by $w/2$ each in opposite directions; the stability of the observer's unit is the constancy of their sum, $\psi + N\psi = 2$; light, being a change of configuration, reads the product of the local units of time and length, $N\psi^2 = \psi \cdot N\psi$.

[C] Then **$N\psi^2 = (1 + w)(1 - w) = 1 - w^2$**: the square of 11.1 follows from a fixed sum read as a product, with no further assumption, and in particular with no independence between the two shares, which are as dependent as two quantities can be: one gives what the other receives. The ratio of the shares, $\psi/(N\psi) = 1/N$, is $1 + z$ for light emitted at the place and received far away. w is the imbalance between the two shares, twice the share transferred; in the ideal Brownian model of 11.3 it is also a probability of arrival.

Status: P8 is a premise and the three identifications are a declared formalisation of it. As in 3.5, the two shares sum to one by construction; P8 reads this as an exchange, and a dynamics of the exchange is not given here. Kept point by point inside matter, this reading gives the closure of section 4.

11.3 The deposit as a probability of arrival. [C] For a Brownian motion in three dimensions, without stopping, the probability of ever reaching a ball of radius ε from a point at distance $r \geq \varepsilon$ is ε/r: the deposit of 3.6, $1/(2\pi r)$, and this probability differ by the constant $2\pi\varepsilon$. Postulate T gives $w = \varepsilon/r$ for every r when **$\varepsilon = GM/(2c^2)$, the horizon radius in isotropic coordinates**; the radius is the coefficient of T (738 m for the Sun, 2.2 mm for the Earth, as virtual targets). With the Euclidean reference metric δ, w is harmonic, equal to 1 on the sphere $r = GM/(2c^2)$ and vanishing at infinity,

hence the probability that a Brownian path reaches that sphere (Kakutani 1944, for regular compact targets in $\mathbb{R}^3$).

In the spatial metric $\psi^4\delta$ the Brownian motion is generated by $\frac{1}{2}\Delta_g$ and w is not harmonic there ($\Delta_g w = 2\psi^{-5}|\nabla w|^2$). The harmonic function of that metric equal to 1 on the horizon and 0 at infinity is $1 - N = 2w/(1 + w)$ (`codice/quota_e_arrivo.py`): **the lapse is the probability that a Brownian path of the spatial geometry escapes to infinity without reaching the horizon.** The lapse of a static vacuum is harmonic in the spatial metric (used, for instance, in the uniqueness theorem of Bunting and Masood-ul-Alam 1987), and the same harmonic function defines the capacity of the horizon in Bray's mass–capacity inequality (Bray 2001). In the Euclidean reading w is a probability of arrival; in the spatial reading the lapse N and its complement $1 - N$ are the probabilities of escaping and of arriving. These statements concern the vacuum exterior, $r \geq GM/(2c^2)$, with the horizon as an absorbing boundary and unlimited time; for the Sun and the Earth the target is virtual and the reading concerns the idealised exterior.

11.4 The fragmentation bench. [C] Whether the fragmentation of a register, the bench of 3.6, reaches a cell with the law of its deposit was tested with predictions hashed before each run (`documenti/ARRIVAL_TESTS.md`; the original registrations and notes, in Italian, are in `documenti/registrations_arrival/`). At distance r a chain has about $1/(r^2 + 1/T)$ readings, so its steps are of order r^2, not small compared with a cell of the 3.6 bench; the geometry was therefore scaled down by four (4 000 000 readings per chain, every reading followed, 3000 chains):

r (scaled bench)	**0.0015**	**0.002**	**0.00275**	**0.00375**	**0.005**	**0.00675**
probability of arrival / Brownian law, original scale	0.925	0.900	0.857	0.809	0.732	0.658
the same, scaled by 1/4, exact path	0.998	0.993	0.976	0.952	0.915	0.904
statistical error	0.011	0.012	0.013	0.014	0.014	0.014

At the scaled bench the probability of arrival is proportional to the deposit within 4.4 % (largest spread of the central values, with the estimator of the registration; constant $0.952 \times 2\pi\varepsilon$); comparing data and Brownian reference over the same shell, after the run, gives 4.5 % and 0.948. At the two smallest radii the deficit is compatible with zero; at the others it is 3 to 6 times smaller than at the original scale, a reduction compatible with a large component due to the discrete steps. The arrivals lost when the path is seen only at block points were measured directly on the same chains: 4.7 % at the smallest radius, 1.3 % at the largest. The registered prediction on the size of this reduction held; the one asking for 0.95 at every radius did not (0.915 and 0.904 at the two largest), a check of the faster block generator against the exact path missed its tolerance narrowly, and at the original scale the registered criteria were not met (`documenti/ARRIVAL_TESTS.md`). The results support the hypothesis of a Brownian limit for the first arrival, which the invariance principle leads one to expect; a proof would need a convergence argument for the paths of the fragmentation. The bench lives in the Fisher space of register states (P3) and stops at a finite clock s_{max}, hence the erfc factors.

12. Anticipated questions

1. **Between which distributions is the overlap, and why not the relative entropy?** Between the Dirichlet posteriors of the place and of the distant observer, projected on the plane perpendicular to the line joining them (3.4); in the many-readings limit they are isotropic, with width $1/\tau$. The normalised overlap is the quantity whose two directions sum to a constant; the relative entropy is not (3.1). Taking the overlap seen from the place is a declared choice.
2. **Isn't this just the linearity of Laplace's equation in vacuum?** Yes. Outside matter the relation is a property of vacuum general relativity; the content is inside matter.
3. **What is the covariant form? From which quantities are ψ and N read?** None yet. In the static spherical domain ψ and N are invariantly defined: N is the norm of the static Killing vector, and ψ^4 is the conformal factor of the spatial metric, which is conformally flat for any spherically symmetric 3-metric.
4. **Why D = 2, if space has three dimensions?** Because the comparison is made on the image perpendicular to the line of sight (premise P2). In the full three-dimensional state space the overlap goes as the 3/2 power (3.4). No apparatus that measures the overlap is identified.
5. **Which equation is dropped, and what about energy conditions?** The lapse equation of general relativity. The Hamiltonian constraint is kept and the pressure follows from conservation, so the fluid's pressure does not enter the field equations. Read in general relativity, the star carries an extra anisotropic stress; above 4/9 it must violate $p_r \geq 0$ or $p_r + 2p_t \leq \rho$ (Andréasson 2008). For the uniform star the fluid pressure is positive and decreasing outwards (`codice/interno_uniforme.py`); energy conditions are examined in MM 2.34.
6. **Why S4 and not S3?** Both give $\gamma = \beta = 1$; S4 reproduces isotropic Schwarzschild at every order, S3 differs at third order. The choice uses general relativity as a criterion; it is not imposed by the premises.
7. **Is the chain circular?** Where Schwarzschild enters, it is said: through Postulate T and the choice S4 (3.7). The

other steps compute forms (a ratio of densities, a share, a 1/r) that do not contain the metric.

13. How to reproduce (Python 3 with numpy, scipy, sympy)

```
python codice/run_all.py                      # all the programs below; logs and JSON in uscite/,
about 90 minutes
python codice/run_all.py --short              # the same without the three long runs of section 11,
about 25 minutes
python codice/ppn_quadrato.py                 # 11.1: gamma and beta for F = 1, 1 - w, 1 - w^2
python codice/einstein_vuoto_esatto.py        # one centre: isotropic Schwarzschild, Einstein tensor
identically zero
python codice/domini_e_masse.py               # two centres: not a vacuum solution; flux = 4 pi
(M_Komar - M_ADM)
python codice/specchio.py                     # psi + N psi = 2 in vacuum; its violation inside
matter in GR
python codice/interno_uniforme.py             # uniform star: C_max in closed form, central
pressure, extra stress
python codice/reciprocita_nello_specchio.py   # 3.1: the overlap identity and the KL comparison
python codice/perielio.py                     # beta = gamma = 1 and integrated exterior orbits
python codice/gamma_anello.py                 # gamma on a finite-count ring
(documenti/NOTE_GAMMA.md)
python codice/ritmo_per_lettura.py            # 3.3: expected Fisher step per count
(documenti/NOTE_FISHER_STEP.md)
python codice/normalizzazioni.py              # 3.7: the four normalisations S1-S4
python codice/grafo_uno_su_r.py [n N seed]    # 6: the free exponent of the graph response (default
n = 512)
```

The programs of the registered tests write their JSON in the current folder; run them from `uscite/` (as `run_all.py` does): `t1_densita.py`, `t5_coerenza.py` (3.3), `t6_specchio.py`, `t9_immagine_esatta.py`, `t9c_controllo.py` (3.4), `t8b_frazionamento.py` (3.6, the longest), `t11_due_masse.py`, `t11b_tensore.py` (3.8); and, for section 10, `angolo_retto.py` (M1-M5: random, grid and fixed-place copies, 1+1 and 3+1, about 45 s), `conteggio_p6.py` (M6: the count with the product rule) and `orologio_fermo_p7.py` (M7: indivisible steps, the clock at rest); for section 11, `cella.py` (first test, 6000 chains), `cella_diagnostica_eps.py 0.004|0.008` (larger cells, after the run), `scala.py A|B 3` (block generator), `valuta_scala.py`, `diag_griglia_A.py 1000` (arrivals lost at block points), `scala_esatta.py 3000 3` (exact path, about 45 minutes on three cores), `valuta_scala_esatta.py` and `quota_e_arrivo.py` (11.2-11.3, symbolic; it reads `scala_esatta.json`). Their comments and printed labels are in Italian.

Reference outputs are in `risultati/`; programs that write files write them to `uscite/`; `run_all.py` compares the two (logs and JSON with line endings ignored; the running times printed by the graph program and by `angolo_retto.py` masked) and exits with an error if an expected output differs or is missing. SHA-256 fingerprints are in `IMPRONTE.sha256`. Code comments are partly in Italian.

14. Glossary

term	meaning here
register	a count vector over K = 4 categories, with its Dirichlet posterior
reading	one count
Bayesian overlap	w_ij of section 2; $\ell(a \to b)$ of section 3 for Gaussians
depth ψ	the conformal factor of the spatial metric
rate N	the lapse
effective density τ	T + 1, the readings of a register plus one
share	$\psi/2$ or $N\psi/2$: the part of the joint count of two points of view held by the place or by the distant observer
fragmentation	the loss of precision of the copies of a register, reading by reading; it behaves as a Brownian motion in Fisher space
Postulate T	the transfer law of section 3.7

copy	in section 10, an event of the count; P7: copies have no pre-assigned places
light sides a, b	the two sides of the light diamond between the start and end of a clock, $c\Delta t \mp \Delta x$
exchange	in section 11, the transfer of fractions of the distant observer's unit between the two shares (P8)
probability of arrival	the probability that a path of the fragmentation, or a Brownian path, reaches a given cell or the horizon
MM	the author's framework (first line)

15. Provenance

The author is not a professional physicist. The framework and its premises are his. The computations were done with AI assistants (Claude and Codex). Predictions of the numbered tests (the author's working series of computations) were hashed before computing in the author's private working files; these hashes are not publicly time-stamped. Checks made afterwards are labelled as such. Some results were re-run by a second AI assistant, not by a human referee. The chain of section 3 was built in a numbered series of registered tests (`documenti/TUNNEL_TESTS.md`) and checked in five rounds by the second assistant; its corrections are applied. Before deposit the whole package was re-run from scratch and every result was reproduced. Version 2.36.1 (October 2026) adds section 9 and its references after a literature check made with the same AI assistant, and section 10 from a separate series of registered tests (M1-M7, `documenti/MOVING_CLOCK_TESTS.md`), with its three programs, which `codice/run_all.py` now also runs; every other program, result and log of version 2.36 is unchanged. Version 2.36.2 (October 2026) adds section 11 from a further series of registered tests on the fragmentation bench (`documenti/ARRIVAL_TESTS.md`), with nine programs, checked by the second assistant before deposit; premise P8 is the author's.

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Sintesi in italiano

- **Fuori dalla materia** $\psi + N\psi = 2$ è una proprietà esatta della metrica di Schwarzschild in coordinate isotrope. Per un solo centro, $\psi = 1 + a/r$ con $N = 2/\psi - 1$ ridà Schwarzschild; con due centri no. Il dominio è statico, sferico, un corpo.
- **Il modello** è definito in cinque righe (sezione 2).
- **Dalle premesse ai due numeri (sezione 3).** Con cinque premesse del quadro (orologio, immagine, distanza, massa, assorbimento): il ritmo del tempo è il rapporto delle densità di letture, $N = \tau_0/\tau_x$; la profondità è la sovrapposizione delle immagini, $\psi = 2\tau_x/(\tau_x + \tau_0)$, nel limite di molte letture; $\psi/2$ e $N\psi/2$ sono le quote del posto e dell'osservatore lontano nel conteggio unito, e $\psi + N\psi = 2$ dice che le due quote sommano a uno. Il frazionamento di un registro si comporta come un moto browniano in tre dimensioni e deposita $1/r$.
- **Un postulato.** Il passaggio dal deposito a ψ è il postulato di trasferimento T ($\psi - 1 = \Sigma\, Gm/(2rc^2)$), scelto fra quattro normalizzazioni perché riproduce esattamente la relatività generale: due sono escluse dal perielio, una differisce solo al terzo ordine. Schwarzschild non è derivata: entra con T. Due corpi fermi danno i dati iniziali di Brill e Lindquist, che non restano statici; la dinamica non c'è.
- **Dominio:** la relazione vale solo in simmetria sferica esatta, un corpo; il grafo del modello vive nello spazio degli stati dei registri, identificato con lo spazio fisico dalla premessa P3.
- **Dentro la materia** è una chiusura fenomenologica delle equazioni statiche, non ancora una teoria: si tiene il vincolo hamiltoniano, la relazione sostituisce l'equazione del lapse, la pressione viene dalla conservazione e non entra nelle equazioni di campo. È la completezza delle due quote, tenuta punto per punto. Letta nella relatività, la stella porta uno sforzo anisotropo in più. La scelta della variabile isotropa ψ è un'assunzione. Il limite è $C_{max} = 2^{(1/3)} - 2^{(-1/3)} \approx 0{,}466$ contro 4/9; a compattezza osservata la pressione centrale è più bassa del 7-11 %.
- **L'1/r sui grafi** è atteso per un Laplaciano di grafo in tre dimensioni: riflette $K = 4$, non una dinamica.
- **Lavori precedenti (sezione 9, aggiunta nella 2.36.1).** Non sono nuovi: la distanza come distinguibilità di Fisher (Wootters; Caticha), il tempo che avanza con l'ampiezza delle fluttuazioni (Caticha; Ipek e Caticha), la forma del lapse come rapporto fra lontano e vicino (Tolman ed Ehrenfest), $-\Phi/2c^2$ come frazione fra 0 e 1, piena sull'orizzonte (Verlinde, nella versione preprint del 2010; l'articolo pubblicato nel 2011 scrive $-2\Phi/c^2$; la quota

w del postulato T coincide con $-\Phi/2c^2$ al primo ordine). Propri di questo deposito, nei lavori esaminati: il lapse come rapporto di densità di letture, con il passo $\sqrt{(K-1)/\tau}$; ψ come sovrapposizione delle immagini e $\psi + N\psi = 2$ come completezza di due quote; il frazionamento browniano che deposita $1/r$; la chiusura dentro la materia.

- **Orologio in moto (sezione 10, aggiunta nella 2.36.1).** Fra partenza e arrivo i raggi di luce disegnano un rombo di lati a e b. Contare le copie dà la media geometrica $\sqrt{(ab)}$, cioè Einstein; sommare i lati dà la media aritmetica, cioè Galileo. Con un orologio che legge la catena più lunga, le copie a caso, senza posti prestabiliti, danno Einstein e una griglia di posti dà Galileo; il criterio complessivo registrato per M1-M5 non è stato soddisfatto (M5 a $v = 0{,}8$, per la taglia). La matematica è nota (insiemi causali, calcolo k di Bondi); il quadro dà la ragione della disposizione. Il legame con il lapse della sezione 3 non è fatto.
- **Scambio e probabilità di arrivo (sezione 11, aggiunta nella 2.36.2).** La relazione $\psi + N\psi = 2$ è letta come la legge di conservazione di uno scambio fra le due quote, che tiene fissa l'unità dell'osservatore lontano (premessa P8); la luce legge il prodotto delle due unità locali e ne esce $1 - w^2$, il quadrato che il secondo ordine richiede, senza ipotesi di indipendenza. Lo sbilanciamento w fra le due quote è, nel modello browniano ideale con il bersaglio fissato da T, la probabilità di raggiungere l'orizzonte isotropo; nella geometria dello spazio il lapse N è la probabilità di sfuggire all'infinito senza raggiungerlo. Sul banco del frazionamento, ridotto di quattro volte, la probabilità di arrivo diventa proporzionale al deposito entro il 4,4 % (stime centrali); la previsione registrata sulla riduzione dello scarto è superata, le soglie più severe (E1, E3) no; i risultati sostengono l'ipotesi del limite browniano.
- **Provenienza:** l'autore non è un fisico di professione; le premesse sono sue; i calcoli sono fatti con assistenti IA; la catena della sezione 3 viene da test registrati prima dei calcoli, controllati in cinque giri da un secondo assistente.